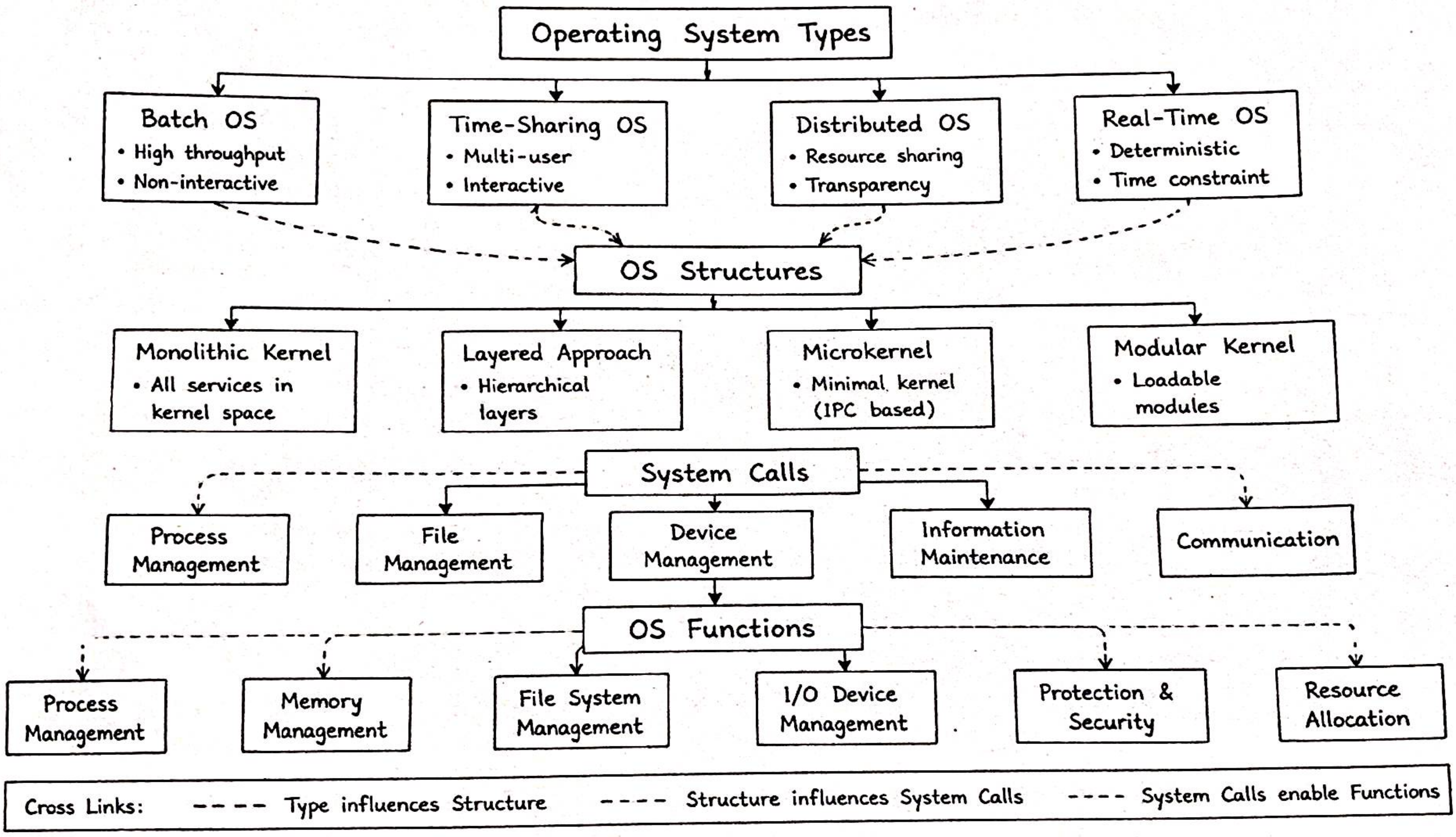
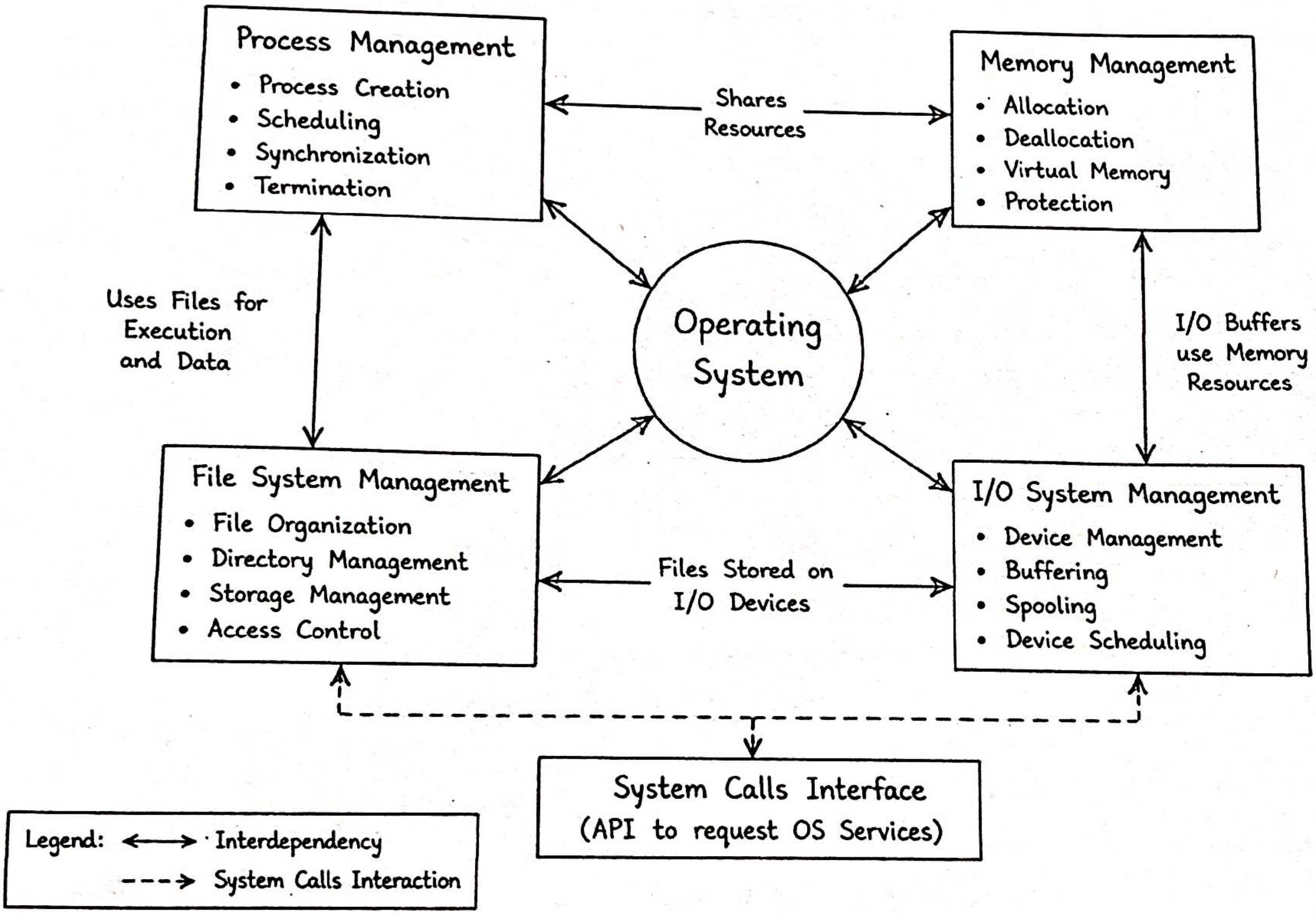
Q1. OS Types → OS Structures → System Calls → OS Functions



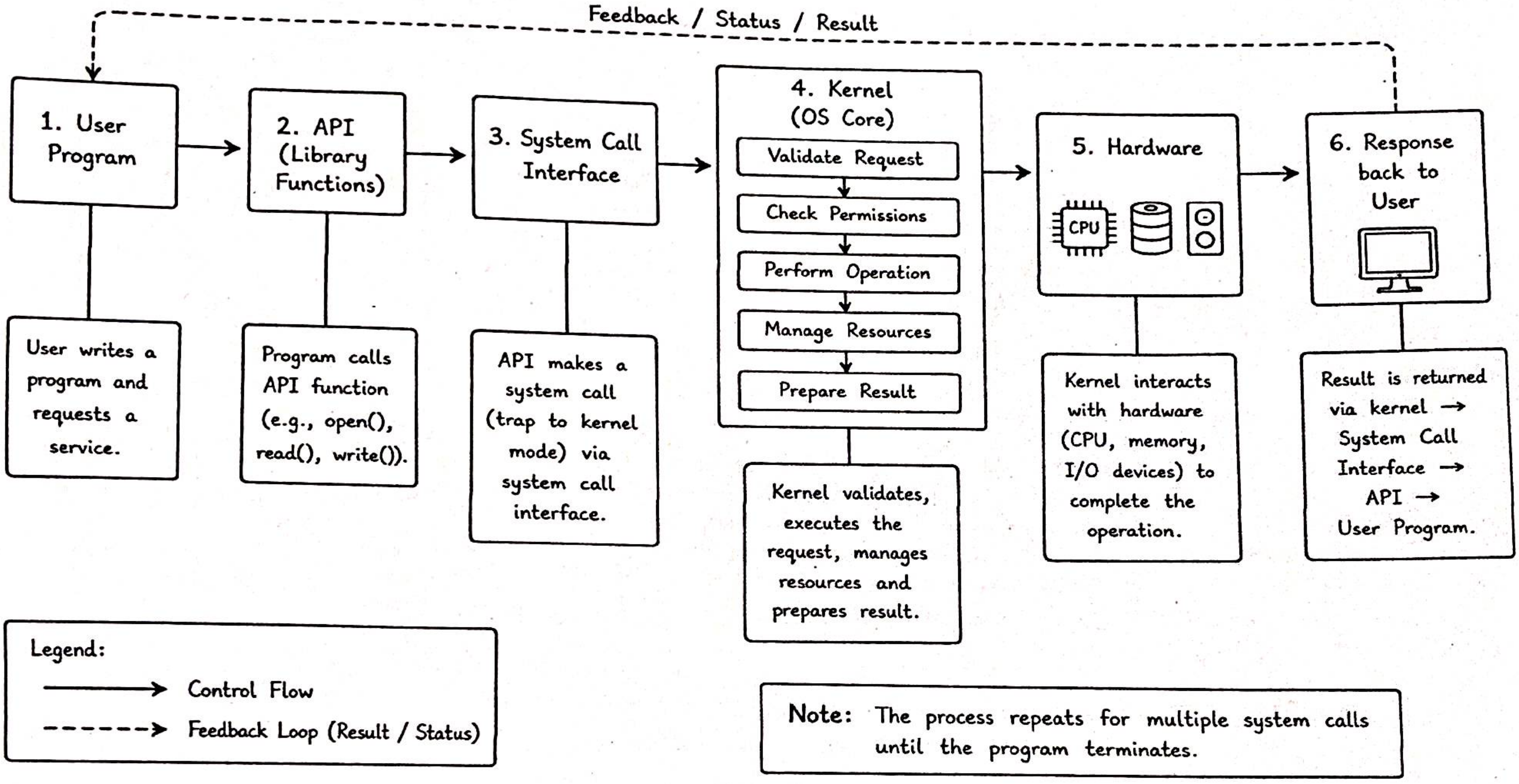
Q2. Process, Memory, File System, I/O System with Interdependencies



Q3. Comparative Concept Map of OS Types

	Batch OS	Time-Sharing OS	Distributed OS	Real-Time OS
Definition	Jobs are collected and executed in batches without user interaction.	CPU time is shared among users interactively.	Manages a group of independent computers and presents as a single system.	Responds to events within strict time constraints.
Key Features	• High throughput • No interaction • Spooling	• Multi-user • Short response time • Time slicing	• Resource sharing • Transparency • Scalability	• Deterministic • Reliability • Low latency
Advantages	• High CPU utilization • Simple	• Better user interaction • Fair CPU sharing	• Load balancing • Fault tolerance • Resource sharing	• Predictable • High reliability • Fast response
Limitations	• No user interaction • Long turnaround time	• Security issues • Complex scheduling	• Complex communication • High cost	• Expensive • Difficult to design
Similarity	All manage resources and provide services through system calls and aim for efficient execution.			

Q4. Execution Flow: User Program → Response back to User



Q5. Hierarchical + Functional Concept Map of System Calls and OS Services